MIND AI × ONTYPOLOGY

AI DRUG DISCOVERY BRAIN

A REASON-ABLE RESEARCH REPLICA

A complete, accurate way to use it

Grounded in Ontypology's Two Types, built on Mind AI



Small-molecule targeted NCE · from target to lead optimization

Type = Type

Space Type × Time Type

Accurate use · Complete method · Illustrated edition

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Before we begin: three conventions and the demonstration declaration

This is a "usable" book. It answers only one question: when a team hands drug R&D to the AI drug R&D brain, how can every output - target card, hit card, SAR table, optimisation plan, verdict document, evolution ledger - be accurate, checkable, and reproducible.

This book takes the Space Type and Time Type of Ontypology as its theoretical basis and Mind AI as its underlying platform, focusing on the R&D scenario of small-molecule targeted new drugs, from target discovery to lead optimisation. The theory itself is not restated here and the platform itself is not introduced here - they are only used. Before starting, establish three conventions.

Convention one: the English word for Type is Type

The English for "Type" is always Type, never Form: TargetType, HitType and the rest are all Types. When you see Type in documents, prompts, or field cards, it refers to the "Type" of Ontypology.

Convention two: the theory is used, not taught

This book uses the three brackets [ ] { } ( ), the intersection criterion ∩(T), the identity anchor id(C), the three levels of remove, the four framework steps, the three-layer judgement, and the fourteen-move tray directly, without arguing again why they hold. For the principles, return to the original Ontypology texts; this book's responsibility is to "use them correctly".

Convention three: the platform is cited, not restated

This book takes Mind AI as its underlying platform: the LLM handles induction and generation in the language layer, the Mind Model handles reasoning in the criteria, deduction, and evolution layers. The platform's architecture and implementation are outside the scope of this book - this book only tells you how to make the output accurate while standing on the platform.

Demonstration declaration

The compound codes (DX series), activity and pharmacokinetic data, and criteria values in the cases of this book are demonstrative, used to illustrate the method, and do not point to any real drug or clinical decision; for any real R&D, follow the experimental data and regulatory requirements. This book involves no system implementation.

Table 0-1 Task map: eight stations, take what you need

Station	Topic (deliberately worded differently from the chapter title)	What you will get
Chapter 1 (understanding the system)	Why a drug R&D brain is needed	The worldview and division of labour of a reasonable R&D twin
Chapter 2 (output Types)	The six-Type relay of target, hit, SAR, optimise, verdict, evolution	Six output-Type field cards and the three-bracket breakdown
Chapter 3 (the four accuracy gates)	How the four gates of accuracy operate	Criteria, derivation, validation, evidence chain
Chapter 4 (six steps)	Operating flow and move selection	Four framework steps, operation card, three-layer judgement
Chapter 5 (prompts and case)	The S1 to S7 seven-stage relay and a full demonstration	Full prompt text + DX series output examples
Chapter 6 (misuse and upgrade)	Control check and the maturity path	Misuse table, ten questions, L1 to L4 blueprint
Review (closing)	Three takeaways	A one-page review chart and a summary table of key points
Appendix A (quick reference)	Terminology, moves, notation	Three quick-reference tables

Guided reading: two languages and three ways of reading

Only two languages meet in the body of this book: the "Type" language of Ontypology and the "on-site"

language of drug R&D. Once you understand their correspondence, the remaining five chapters are only elaboration.

Table D-1 Two languages compared

The language of Ontypology (tools)	The language of drug R&D (on site)
Type	Target, molecule, series, verdict - anything can be established as a Type
Property [ ] (abstract set)	The scales of activity, selectivity, druggability, and ADMET
Composition { } (composition set)	Scaffold to functional group to atom: the dependency structure
Container ( ) (container set)	R&D programme to compound series to the space of a single molecule
Intersection criterion ∩(T)	Pass lines and veto lines of the four gates: the generation rule of qualified output
Identity anchor id(C)	The stable identity of molecule and series: renamed or re-versioned, it still recognises itself
Moves (Time Type)	Methods of search, reasoning, design, learning, and evolution
Evolution ledger	Each round's record, attribution, and plan for the next round

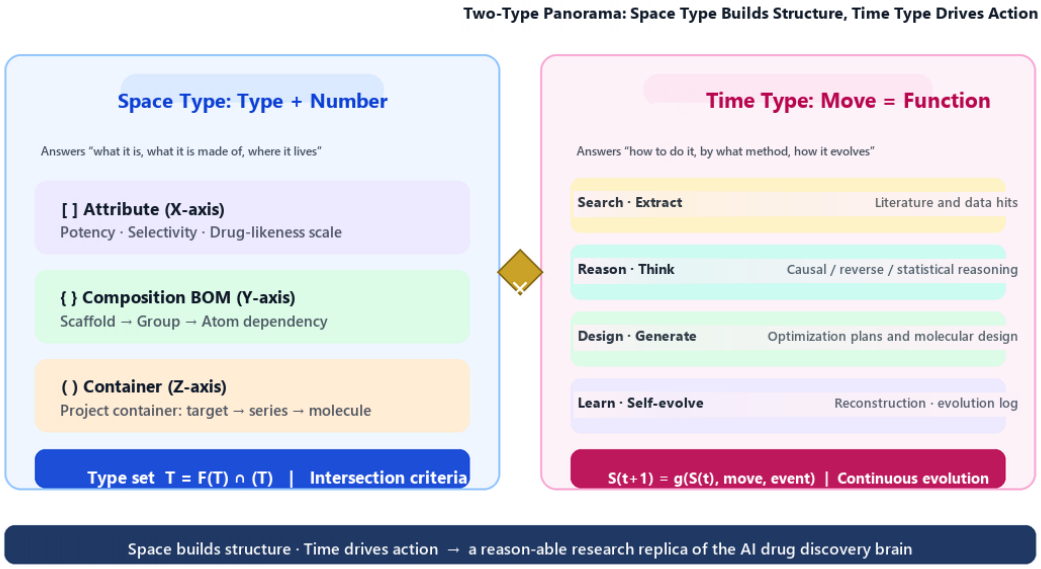
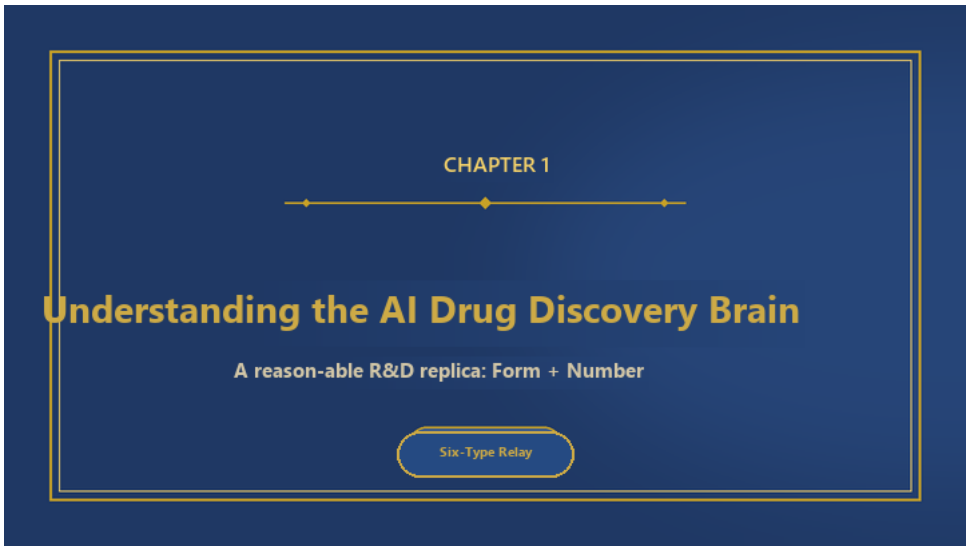


Figure D-1 Panorama of the two Types: the Space Type establishes structure, the Time Type establishes action

Table D-2 Three ways of reading

Reader	How to read it	Where to stop reading
Decision maker	Guided reading, the orientation chapter, and the misuse and maturity chapter	Know what this brain can give you
R&D director	Add a reading of the output-Type chapter and the four-gate accuracy chapter	Can establish Types, can accept, can run a review meeting
Executing colleague	Read the whole book straight through; copy the prompt chapter	Start directly at S1 to S7

Chapter 1 Understanding the AI Drug R&D Brain



The reasonable drug R&D twin: Type + Number

The most expensive thing in drug R&D is not a compound but a decision made with no idea why: why is this hit worth pursuing? Why should this lead be changed this way? Why dare to send this candidate to the next gate? The reason the AI drug R&D brain exists is to establish the R&D site as a reasonable twin, so that every output can say "what it is, on what grounds, and how it came about".

1.1 A Reasonable R&D Twin: Type + Number

A twin is not a database. A database answers "how much"; a twin must also answer "what it is" - that is Type + Number: Type writes structure, composition, container, and criteria; Number writes the values of activity, property, and relationships. Only when an R&D programme is established as Type + Number can AI ask "what it is" before it opens its mouth and produce evidence before it draws a conclusion.

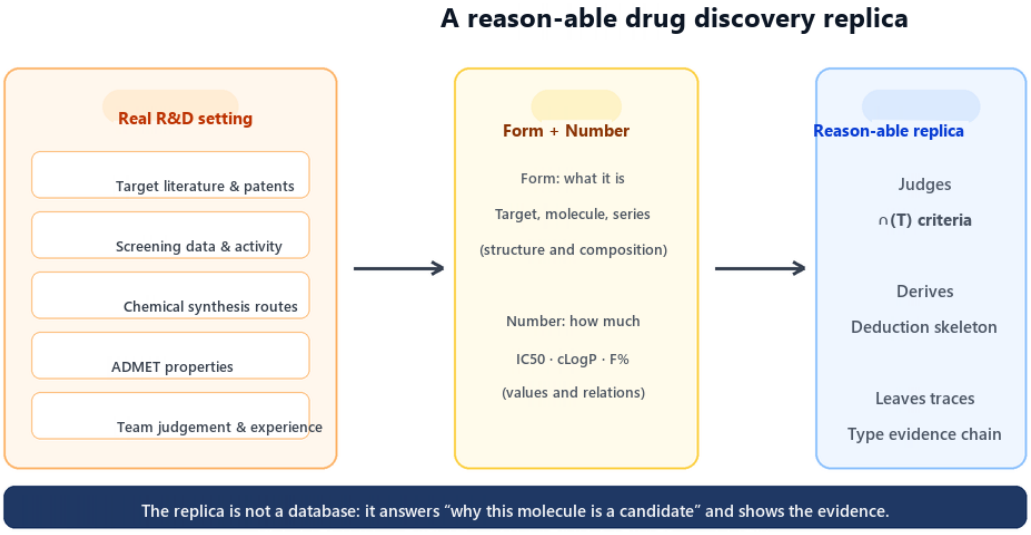


Figure 1-1 A reasonable drug R&D twin

> **Example: the Type and Number of DX-201**
> Type: it is a member of the "phenylpyrimidine series" under the "fourth-generation EGFR inhibitor programme" container, and the scaffold is a pyrimidine-aniline bicyclic.
> Number: EGFR T790M/C797S IC50 = 2.4 μM, LE = 0.31, cLogP = 2.8.
> Only when both sides are put together is the Type "established" - report only numbers and nobody can catch the next round.

1.2 The Two Types Land in Drug R&D

The Space Type answers "what it is, what it is made of, where it lives": the property [ ] is the scale axis of activity and druggability, the composition { } is the dependency axis from scaffold to atom, the container ( ) is the spatial axis from programme to molecule. The Time Type answers "how it is done, how it evolves": search, reasoning, design, and learning are all moves, written into the evolution ledger round after round. Where the two Types meet is where R&D decisions happen.

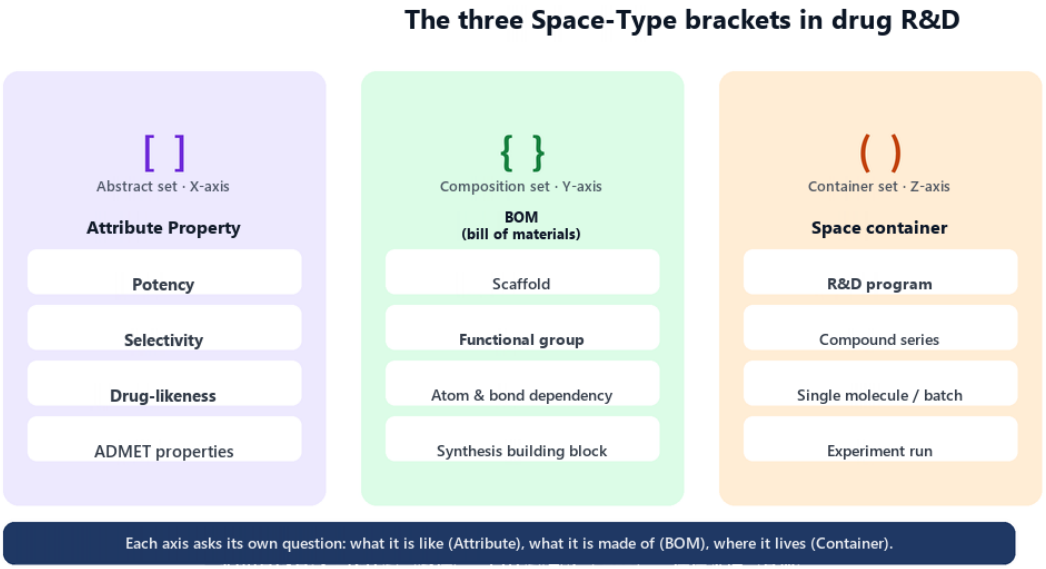


Figure 1-2 The three brackets of the Space Type land in drug R&D

Table 1-2 Where the three Space brackets land

Bracket	Ontypology tools	Where it lands in drug R&D
[ ] Property	Abstract set, X axis	The scale card of activity, selectivity, druggability, and ADMET
{ } Composition	BOM composition set, Y axis	Scaffold, functional group, atoms, and synthesis building blocks
( ) Container	Container set, Z axis	R&D programme to compound series to single-molecule batch
∩(T) criterion	Intersection criterion	Four-gate pass lines: activity / selectivity / druggability / ADME
id(C) anchor	Identity anchoring	Renamed or series-upgraded, it still recognises itself

1.3 The Four Sets of the Move Tray and Where the Fourteen Moves Land

The fourteen moves of the Time Type fall into four sets: search and extraction, reasoning and thinking, design and generation, learning and evolution. Every step of R&D selects a move - target exploration uses search and extraction, SAR explanation uses causal reasoning, molecular design uses design patterns, and each round's closing uses self-evolution. A move is a verb and the output is a noun; pick the right verb first and the noun will be accurate.

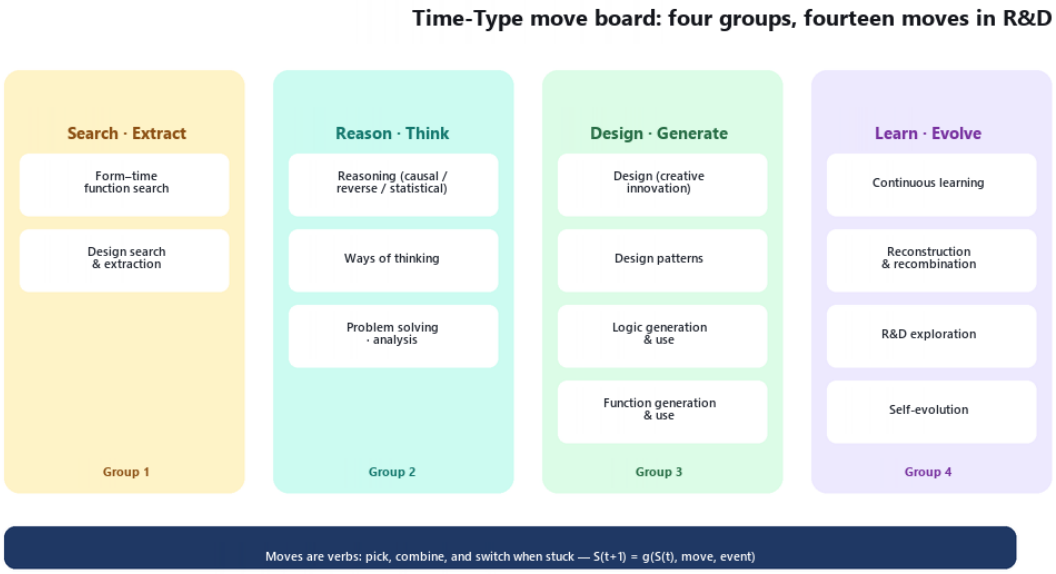


Figure 1-3 The four sets and fourteen moves of the Time Type move tray

Table 1-3 Where the fourteen moves land in drug R&D

Set	Move	Where it lands in R&D	Output verb
One	Type-time function search	Hits from literature, patents, databases	Find: who has solved this problem
One	Design search and extraction	Virtual library screening, fragment screening	Screen: which molecules have a chance
Two	Reasoning (causal, reverse, statistical)	SAR attribution, metabolic site inference	Explain: why the activity changed
Two	Thinking methods	Look at the skeleton with first principles, look at failure in reverse	Think: look from another angle
Two	Problem solving	Untying conflicting parameters	Disassemble: which knot to untie first
Two	Analysis	Binding mode, ADMET alert analysis	Analyse: what the data is saying
Three	Design (creation and innovation)	An entirely new scaffold and design of modification sites	Create: what the next round's molecule looks like
Three	Design patterns	Bioisostere and prodrug strategies applied	Set: mature plays used directly
Three	Logic generation and application	Retrosynthetic reasoning, route feasibility	Derive: can this molecule be made
Three	Function generation and application	Generative molecular design, property optimisation	Generate: give me a batch of new candidates
Four	Continuous learning	Each round's data feed back into the criteria	Learn: criteria grow with the data
Four	Refactoring and recombination	Scaffold hopping, fragment recombination	Restart: escape the local optimum
Four	R&D and exploration	Exploration of new targets and new indications	Explore: where the next battlefield is
Four	Self-evolution	Evolution ledger to next round's plan	Optimise: mistakes never overnight, the next round more accurate

1.4 The Mind AI Division of Labour and the Four Gates

The division of labour at the base is simple: the LLM excels at inductive statistics in the language layer - literature extraction, hypothesis generation, report writing; the Mind Model excels at deductive reasoning in the criteria layer - ∩(T) verdicts, skeleton derivation, remove counterfactuals, id(C) anchoring. Language to the LLM, criteria to the Mind Model; neither crosses the line, and only then is accuracy grounded.

Table 1-4 The Mind AI division of labour

Work	LLM (language layer)	Mind Model (criteria layer)
Knowledge	Literature extraction, abstract, translation	Field-based Type and three-bracket modelling
Reasoning	Inductive statistics, hypothesis generation	Deductive derivation, causal verdict
Output	Description, report, prompt	Criteria enforcement, evidence-chain verification
Memory	Conversation context	Evolution ledger, identity anchor, criteria version

Table 1-5 Preview of the four gates

Gate	One sentence	Who guards it
First layer	Whether it qualifies is decided by the criteria	The four-gate funnel of the intersection criterion ∩(T)
Second layer	Whether it can be derived is decided by the skeleton	Deductive skeleton: premise, rule, prediction
Third layer	Whether it is real is decided by the experiment	The three-piece set: activity, ADMET, safety window
Fourth layer	Whether it can be found out is decided by the evidence	The six stations of the Type evidence chain

1.5 Four Scenes of Inaccurate Output

An accurate method grows out of inaccurate scenes. R&D output is usually untrustworthy because it falls into four scenes - the four gates later in this book are the countermeasures to these four scenes, one by one.

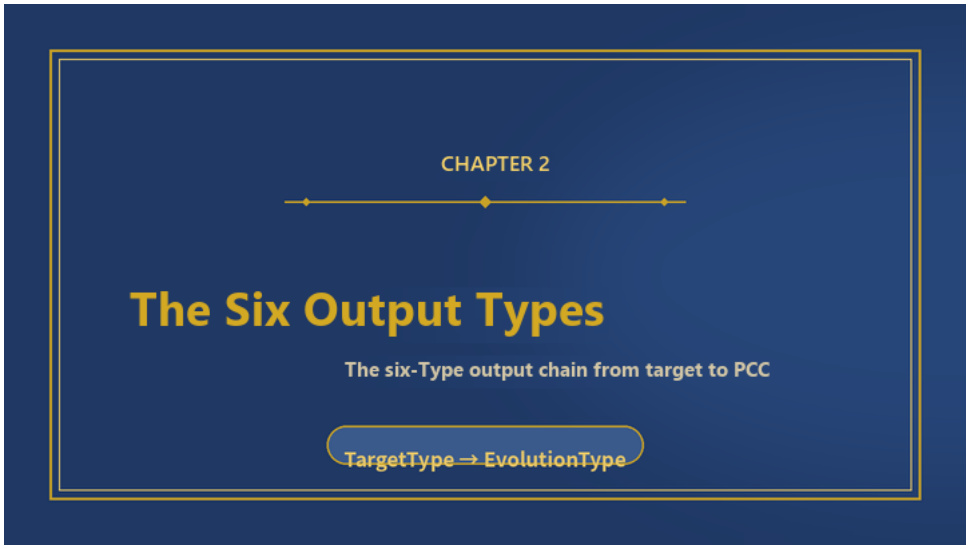
Table 1-1 Four scenes of inaccurate output

Scene	Symptom	Countermeasure (chapter of this book)
The Type is not established	Molecules and series are recognised by nickname, and renaming loses the memory	Six-Type field cards + id(C) (Chapter 2)
Derivation break	Jumping from "chemical intuition" to a conclusion, with no way to reproduce it	The three-piece set of the deductive skeleton (Chapter 3)
Validation drifts	Reporting only the best run, with the raw data missing	The three-piece set of experimental validation (Chapter 3)
Review amnesia	Falling into the same pit once a quarter	Evolution ledger (Chapter 5)

Chapter focus

A reasonable twin = Type + Number: Type writes "what it is", Number writes "how much"; neither may be missing.
Where the two Types land: the three Space brackets establish structure, the fourteen moves establish action, and the evolution ledger turns mistakes into assets.
Iron rule of the division of labour: language to the LLM, criteria to the Mind Model; the four gates are the trailer for accuracy.

Chapter 2 Six Output Types: Target, Hit, SAR, Optimise, Verdict, Evolution



The six-Type output chain from target to PCC

An R&D team produces many documents every day, but only six outputs actually drive decisions: Target Type, Hit Type, SAR Type, Optimize Type, Verdict Type, Evolution Type - remembered as "target, hit, SAR, optimise, verdict, evolution". Six Types in relay, each with a field card, criteria, and evidence; miss one and the chain breaks.

2.1 Six Types at a Glance: the Output Chain from Target to PCC

The six Types relay along the R&D timeline: the Target Type answers "what to attack", the Hit Type "what was found", the SAR Type "what the relationship is", the Optimize Type "what to change", the Verdict Type "does it work", and the Evolution Type "what to change next". The first three belong to discovery and the last three to optimisation - the boundary of this book is the PCC (drug candidate) verdict and no further.

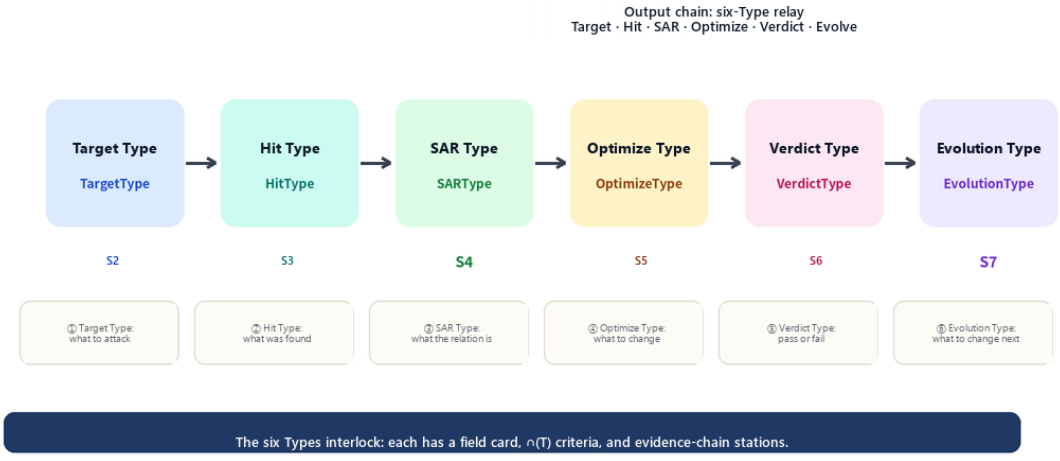


Figure 2-1 The output chain: six Types in relay

Table 2-1 The six Types at a glance

Output Type	English	One sentence	Criteria focus
Target Type	(TargetType)	What to attack: building the target and indication card	Target evidence, druggability
Hit Type	(HitType)	What was found: building the hits-and-screening card	Activity threshold, reproducibility
SAR Type	(SARType)	What is the relationship: building the change-to-consequence card	The rule holds, derivability

Output Type	English	One sentence	Criteria focus
Optimize Type	（OptimizeType）	What to change: building the multi-parameter plan card	Six axes never regress, synthetically feasible
Verdict Type	（VerdictType）	Does it work: building the PCC verdict card	All four gates passed, no break points in the chain
Evolution Type	（EvolutionType）	What to change next round: building the evolution ledger card	Attribution holds, plan executable

Table 2-8 Three levels of remove x compounds: recognise itself after disassembly

Grade	What to disassemble	Judgement and consequence
Level one: key part	Pharmacophore (hydrogen-bond donor and acceptor)	Removing it zeroes the activity - remove(p1, DX-288) is not equal to DX-288
Level two: scaffold dependency	Core scaffold	Take away the scaffold and the series identity (id(C)) changes - it is a new series, not a new molecule
Level three: decorative part	Solubility-modifying group	Removing it lowers the property but identity is unchanged - a new molecule in the same series

2.2 Target Type TargetType

The Target Type is the anchor of the whole chain. It writes "what to attack" as a Type: target name, target variant, indication, mechanism, criteria, and evidence source. If the Target Type is not established, the remaining five Types build on sand.

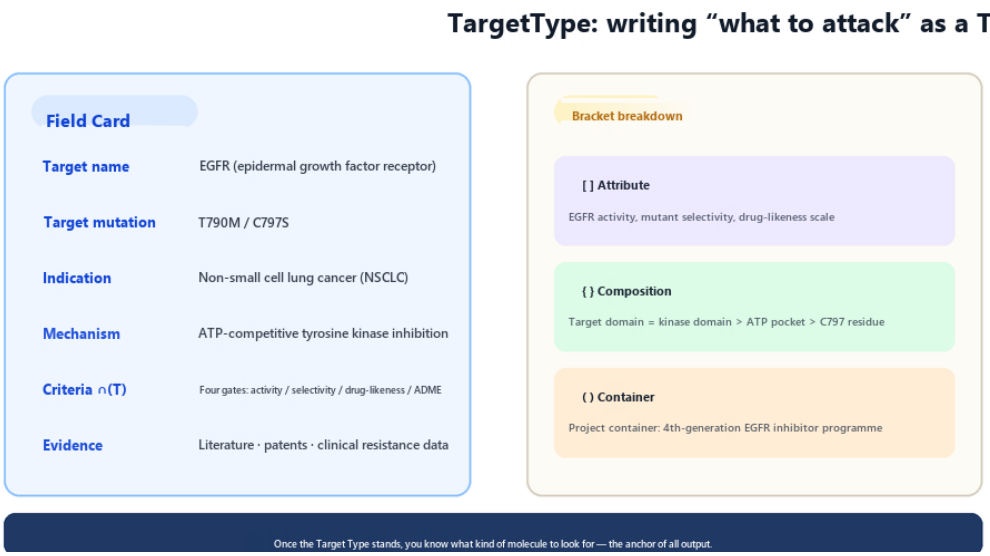


Figure 2-2 Target Type field card

Table 2-2 Target Type (TargetType) field card

Field	What to fill in	Criteria
Target name	EGFR (epidermal growth factor receptor)	Standard naming, alias comparison
Target variant	T790M / C797S cis double mutation	The variant distribution has clinical evidence
Indication	Non-small cell lung cancer (NSCLC)	Indication compatible with the mechanism
Mechanism	ATP-competitive tyrosine kinase inhibition	The binding mode can be verified structurally
Criteria ∩(T)	Four gates: activity / selectivity / druggability / ADME	Set first, use after; any change of line leaves a trace
Evidence source	Literature, patents, clinical resistance data	Traceable, with timestamps

2.3 Hit Type HitType

The Hit Type writes "what was found" as a Type: the structure, activity, how it was found, and the reproduction record of the hit molecule. The greatest taboo of the Hit Type is "reporting only the IC50" - the screening path, the controls, and the replication count decide whether this hit is a lead or noise.

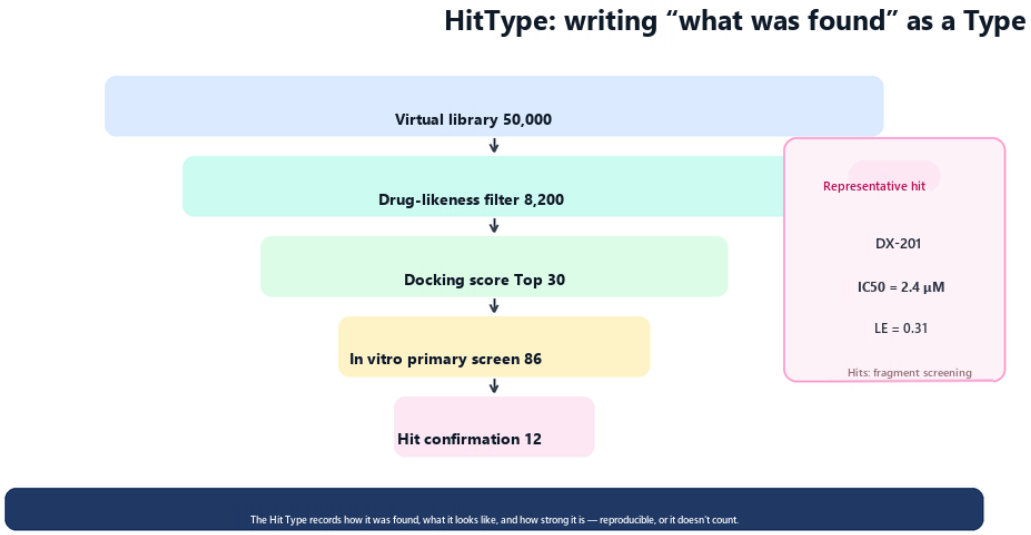


Figure 2-3 Hit screening funnel

Table 2-3 Hit Type (HitType) field card

Field	What to fill in	Criteria
Hit molecule	DX-201 (phenylpyrimidine bicyclic)	Structure confirmed, purity >= 95%
How it was found	Fragment screen	The screening path is reproducible
Initial activity	IC50 = 2.4 μM (mutant)	n = 3 · R^2 ≥ 0.9
Ligand efficiency	LE = 0.31	Only LE >= 0.3 may be shortlisted
Initial selectivity value	WT/mutant = 18	A trend is enough; fill it in during optimisation
Exclude noise	PAINS negative, no aggregation	Noise exclusion complete

2.4 SAR Type SARType

The SAR Type writes "the relationship" as a Type: what was changed, by how much, and because of what. The SAR table is not a report card but a derivable causal record - every column must be able to answer "what happens if it is changed back" (remove counterfactual), and every row must connect to the derivation skeleton.

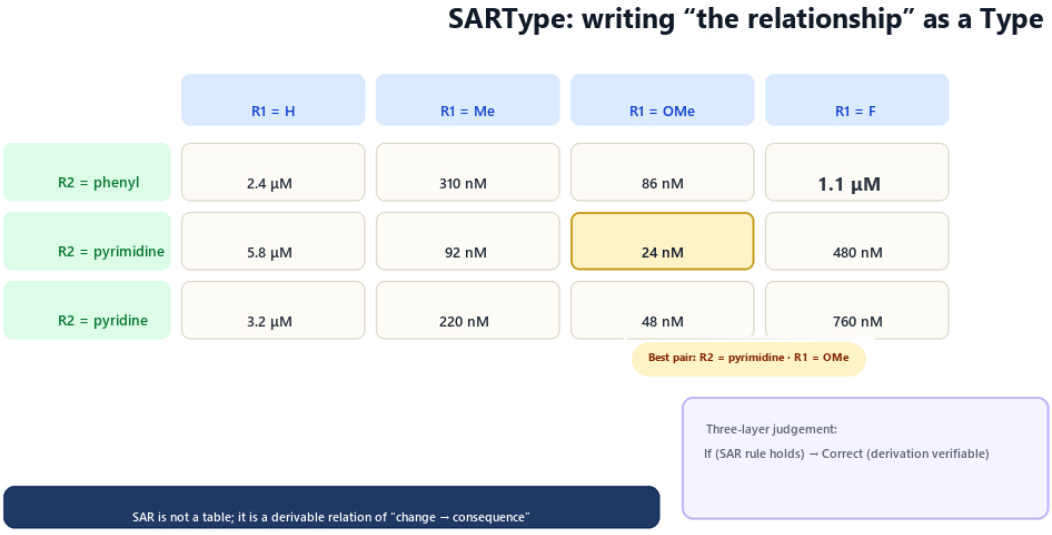


Figure 2-4 SAR matrix: the activity comparison of R1 x R2

Table 2-4 SAR Type (SARType) field card

Field	What to fill in	Criteria
Control axis	R1 = H/Me/OMe/F, R2 = phenyl/pyrimidine/pyridine	Move only one position at a time

Field	What to fill in	Criteria
Effect description	R1=OMe raises electron density and affinity; R2=pyrimidine forms a second hydrogen bond	The mechanism explanation is verifiable
Counterfactual	R1 back to H: activity falls back to the micromolar level (one of the three levels of remove)	Recognise itself after disassembly
Derivation prediction	R2=pyrimidine + R1=OMe estimated at 20-30 nM	Prediction first, measurement after
Measured correction	Measured 24 nM (DX-268)	Differences recorded in the evolution ledger

2.5 Optimize Type OptimizeType

The Optimize Type writes "what to change" as a Type: around multi-parameter optimisation (MPO), it lists the knobs to move this round, the constraints, and the expectations. The bottom line of the Optimize Type is that the six axes never regress - activity gains must not be traded against metabolism, safety, or solubility.

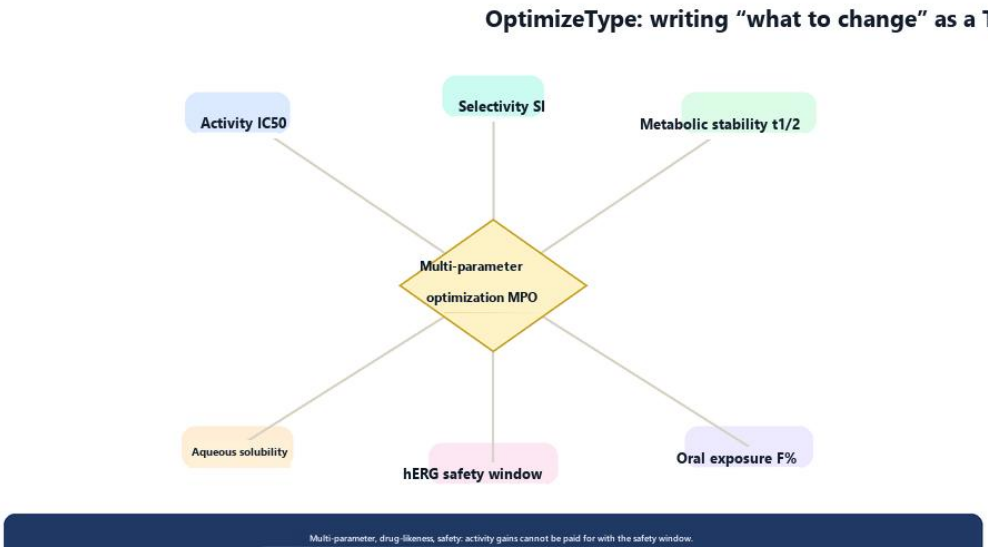


Figure 2-5 Multi-parameter optimisation: all six axes together

Table 2-5 Optimize Type (OptimizeType) field card

Field	What to fill in	Criteria
This round's goal	Metabolic stability: microsomal t1/2 from 18 to >= 20 min	Quantifiable target
Constraints not moved	Activity kept <= 30 nM, SI >= 50	Six axes never regress
Design knobs	Metabolic site blocking + minor polar group adjustment	Synthetically feasible (<= 5 steps)
Expected interval	IC50 10–25 nM · t1/2 25–45 min	Prediction first, with a tolerance interval
Exit conditions	Two rounds without progress, so change the scaffold (refactoring move)	Move-swap conditions written in advance

2.6 Verdict Type VerdictType

The Verdict Type writes "does it work" as a Type: the four gates judged item by item, the evidence chain checked, the conclusion and sign-off. The Verdict Type is an external document - it writes only "pass" and "fail" plus evidence, not mood.

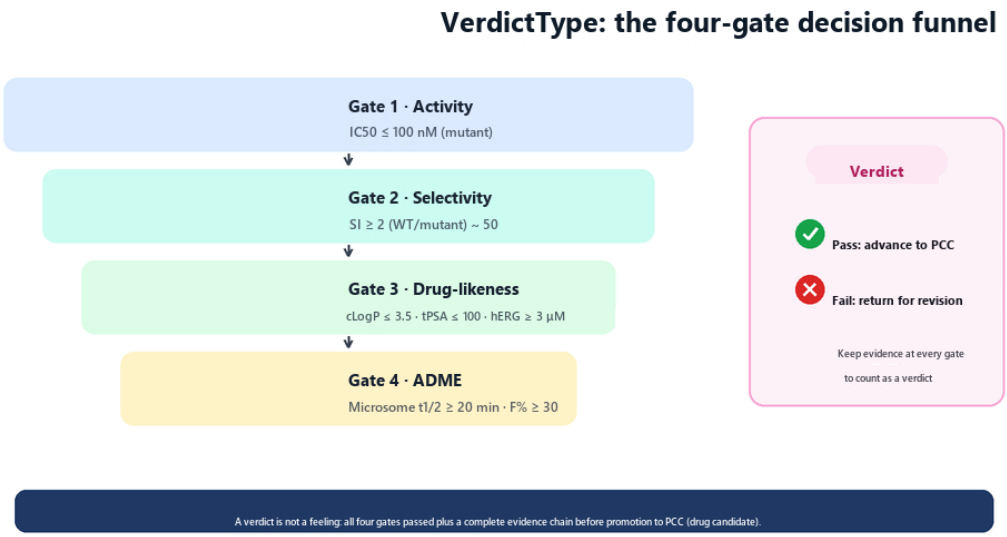


Figure 2-6 The four-gate verdict funnel

Table 2-6 Verdict Type (VerdictType) field card

Field	What to fill in	Criteria
Verdict subject	DX-288 (output of Optimize Type round 3)	Identity anchor id(C) matches
Four-gate results	Activity check / selectivity check / druggability check / ADME check	Item by item with data, no bundling
Evidence chain	All six stations connected (establish Types to archive)	A broken chain means rollback; no verdict may be given
Conclusion	Holds: upgrade to PCC	The conclusion can only be one of three: upgrade, return, shelve
Sign-off	R&D lead, date, criteria version	Person, time, and version: all three present

2.7 Evolution Type EvolutionType

The Evolution Type writes "what to change next round" as a Type: what was changed this round, how much the result differed, what the attribution is, and what to do next round. The evolution ledger is a letter to the next round's self - without accounts, mistakes become habits; with accounts, mistakes become criteria.

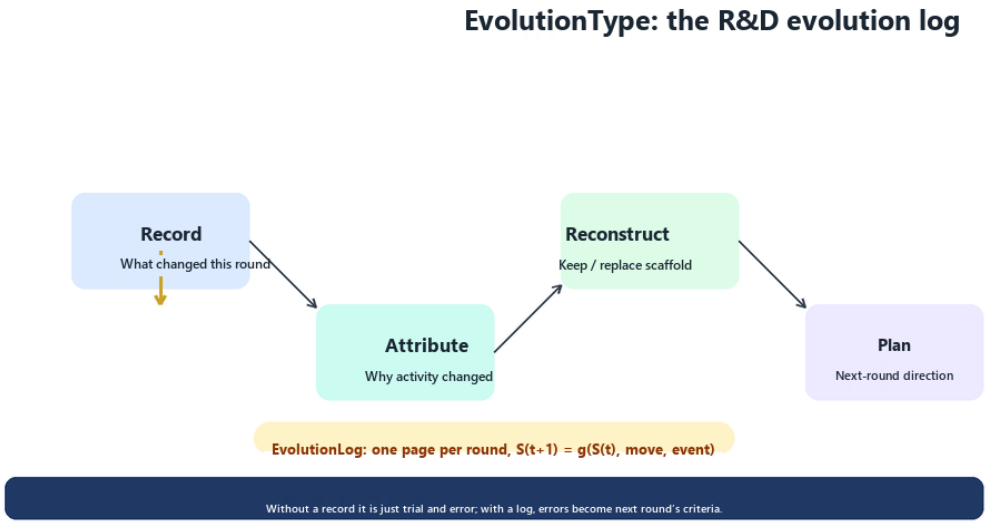


Figure 2-7 The evolution ledger cycle: record, attribute, refactor, plan

Table 2-7 Evolution Type (EvolutionType) field card

Field	What to fill in	Criteria
This round's changes	DX-268 to DX-288: block the 3-position metabolic site	Lines up with the Optimize Type goal

Field	What to fill in	Criteria
Result change	t1/2 18→42 min · IC50 24→12 nM	Speak with data, do not pick the pretty ones
Attribution hypothesis	The metabolic soft spot is blocked and clearance falls	The attribution can be tested in the next round
Criteria update	"Exposed 3-position = metabolic soft spot" enters the criteria library	Learning lands in the criteria, not in the head
Next round's plan	Advance IND development: toxicology and formulation planning	Plan executable, with a schedule

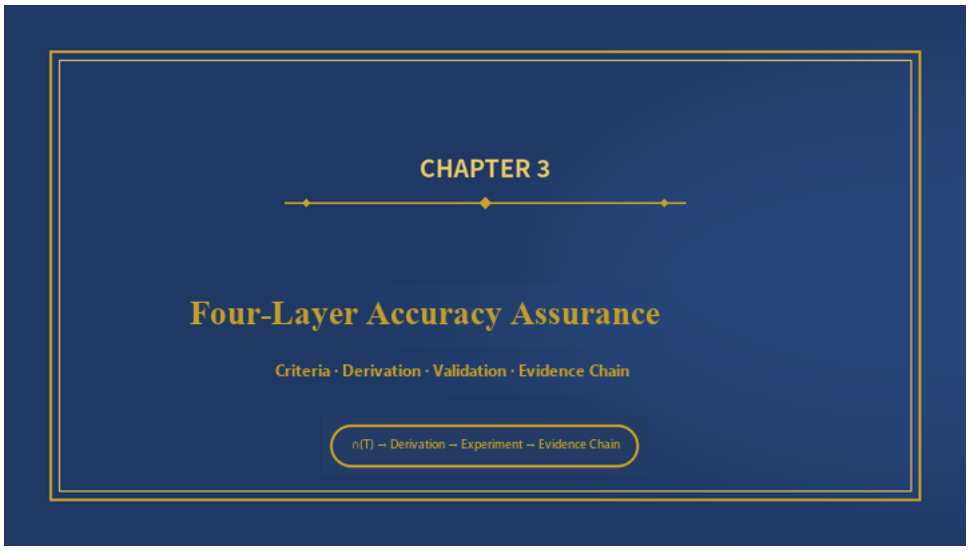
Chapter focus

The six Types "target, hit, SAR, optimise, verdict, evolution" are the minimal complete set of R&D output - one field card per Type, and the fields on the card are the entry points of the criteria.

The three levels of remove let a molecule "recognise itself after disassembly": the key part defines activity, the scaffold defines identity, the decoration defines property.

The Verdict Type writes only evidence and the evolution ledger writes only facts and attributions - these two closing documents determine the starting point of the next round.

Chapter 3 Four-Layer Accuracy Assurance



Criteria, derivation, validation, evidence chain

"Accurate" cannot rely on feeling; it must rely on gates. The four gates take turns: the intersection criterion governs qualification, the deductive skeleton governs reasoning, experimental validation governs reality, the Type evidence chain governs audit; finally Space × Time cross-acceptance performs the overall check. Only when all four layers pass is the output called accurate.

3.1 Layer One: the Intersection Criterion ∩(T)

∩(T) is the intersection of Type and Type: a molecule must both be "a Type of this series" and pass all criteria to count as qualified output. The criteria are divided into four gates - activity, selectivity, druggability, ADME - set first and used afterwards; to change a line, a trace must be left.

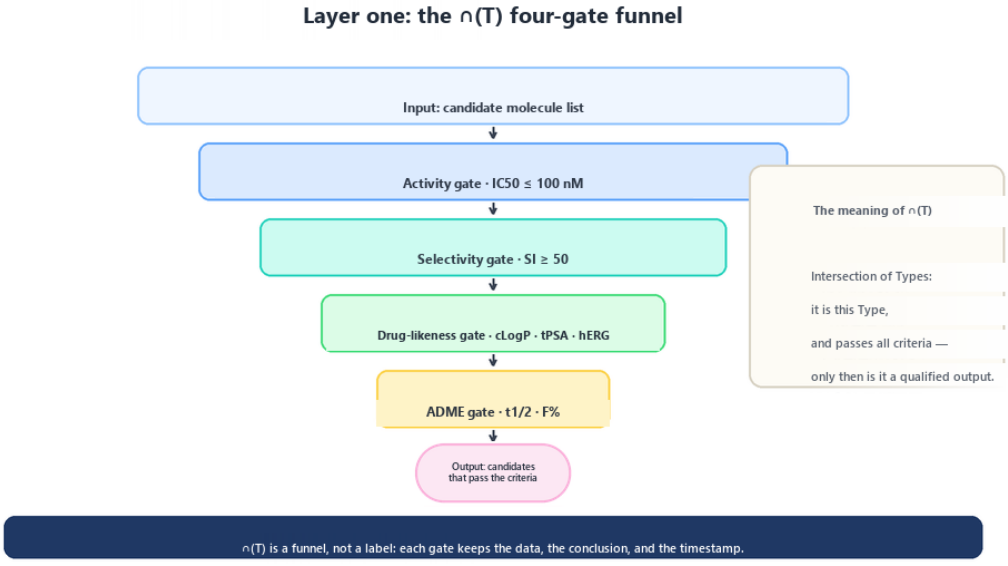


Figure 3-1 The four-gate funnel of the intersection criterion

Table 3-1 The four-gate criteria (DX case version)

Gate	Measurement	Pass line	Veto line
Activity gate	IC50（EGFR T790M/C797S）	≤ 100 nM	> 300 nM: reject outright
Selectivity gate	SI = IC50(WT) / IC50(mutant)	≥ 50	< 20: reject outright
Druggability gate	cLogP · tPSA · hERG	≤ 3.5 · ≤ 100 · ≥ 3 μM	Any red line means reject
ADME gate	Microsomal t1/2, oral F%	≥ 20 min · ≥ 30%	t1/2 < 10 min: reject

3.2 Layer Two: Deriving the Deductive Skeleton

The second layer governs reasoning: every prediction must complete the three-piece set "premise, rule, prediction" before it may land as a conclusion. The LLM induces a hypothesis, the Mind Model derives a prediction and a tolerance interval; once the measurement arrives, correction is made and the difference is written into the evolution ledger. Conclusions that skip steps are stopped right here.

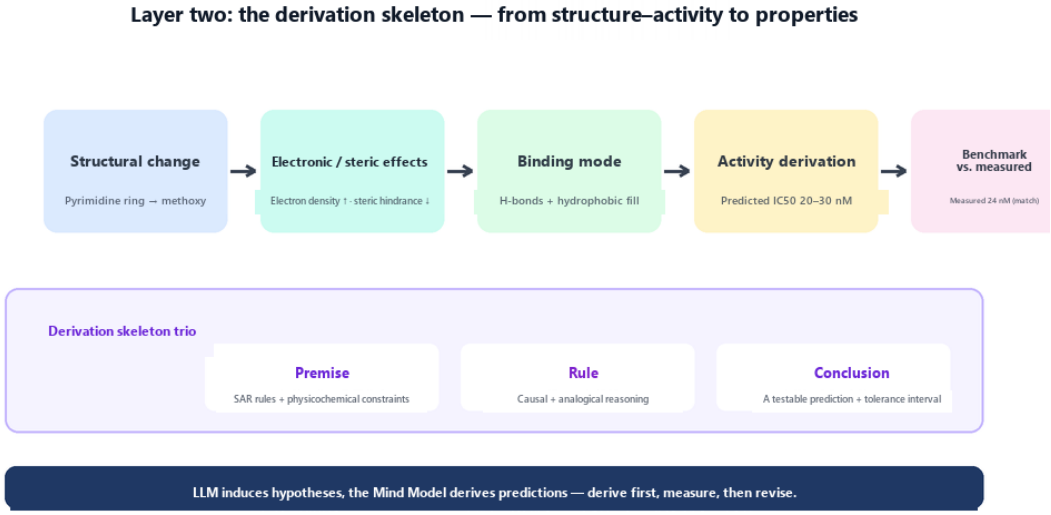


Figure 3-2 The deductive skeleton: from structural change to measured comparison

Table 3-2 Example of deductive skeleton derivation (DX-268 to DX-288)

Step	Content
Premise	DX-268 has activity of 24 nM, but human liver microsomal t1/2 is only 18 min - the ADME gate is not passed
Rule	Causal reasoning: the exposed 3-position of the phenyl ring is a metabolic soft spot; reverse reasoning: blocking the soft spot usually extends t1/2
Prediction	After blocking, t1/2 is estimated at 25-45 min and IC50 stays in the 10-30 nM range
Measured	DX-288: t1/2 = 42 min, IC50 = 12 nM (both within the predicted interval)
Correction	Activity better than the lower prediction bound - the "blocking or affinity-enhancing" hypothesis enters the evolution ledger pending verification

3.3 Layer Three: the Three-Piece Set of Experimental Validation

The third layer governs reality: activity, ADMET, safety window - the three-piece set, each with controls, replication, and statistics. The derivation says "it should be 24 nM", the experiment answers "it really is 24 nM" - where the difference lies must also be recorded.

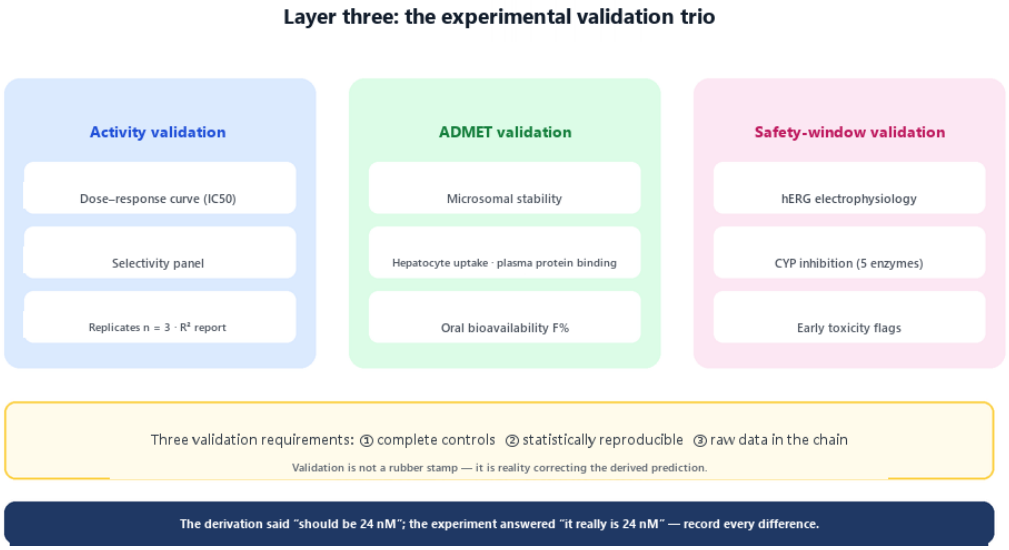


Figure 3-3 The three-piece set of experimental validation

Table 3-3 The validation three-piece set and DX-288 measurements

Kit	Item	Standard	DX-288 measured
Activity	Dose-response IC50 (mutant)	≤ 100 nM · n = 3 · R^2 ≥ 0.9	12 nM（R^2 = 0.99）
Activity	Selectivity panel (WT control)	SI ≥ 50	SI = 118（WT 1.42 μM）
ADMET	Human liver microsomal t1/2	≥ 20 min	42 min
ADMET	Oral bioavailability (mouse)	F ≥ 30%	48%
Safety window	hERG IC50	≥ 3 μM	8.6 μM
Safety window	CYP inhibition (five enzymes)	IC50 ≥ 10 μM	All five enzymes >= 10 μM

3.4 Layer Four: the Six Stations of the Type Evidence Chain

The fourth layer governs audit: establishing the Types, output, derivation, validation, verdict, archiving - each of the six stations keeps three things: a field snapshot, the numerical evidence, and a timestamp. If any station's chain breaks, the output may not enter the next round; a broken chain means rollback and evidence reinforcement, with no exceptions.

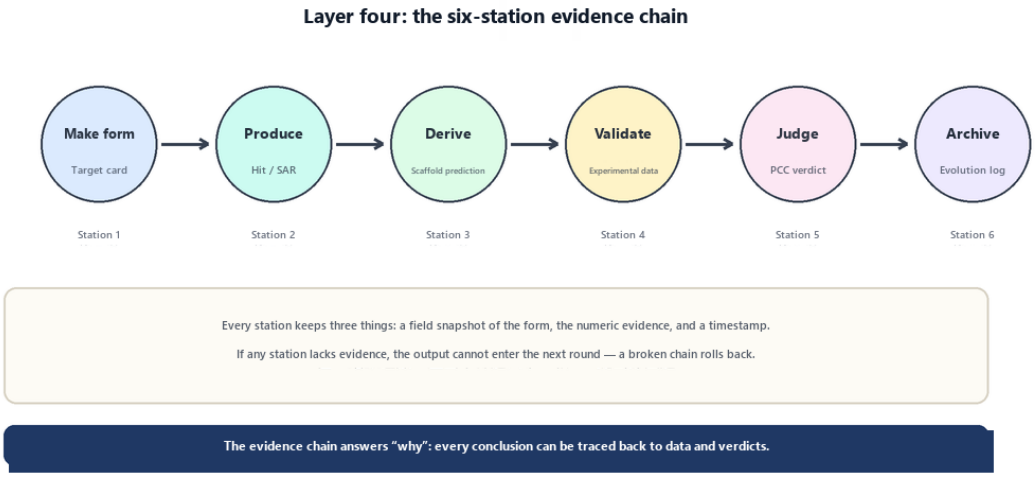


Figure 3-4 The six stations of the Type evidence chain

Table 3-4 The six stations' duties and the consequences of a broken chain

Station	What to keep	Consequence of a broken chain
1 Establish the Types	Target card snapshot + criteria version	Output without an anchor, return to the situation and re-establish
2 Output	Field card + raw data	Not reproducible: void and rerun
3 Derivation	Premise, rule, prediction + tolerance interval	Skipped-step conclusion, return and fill in the derivation

Station	What to keep	Consequence of a broken chain
4 Validation	Raw curves, spectra, statistics	No raw file, no verdict
5 Verdict	Four-gate results + the three sign-off elements	No sign-off means no verdict
6 Archive	Evolution ledger page + criteria changes	Not archiving means this round was wasted

3.5 Space × Time Cross-Acceptance

The final overall check: the three Space questions (Types complete, numbers trustworthy, chain traceable) x the three Time questions (moves aligned, derivation complete, evolution leaves a trace); only when all six pass is it released. Space says "what it is", Time says "how it comes about" - only when both sides match does accuracy hold.



Figure 3-5 Space x Time cross-acceptance

Table 3-5 The six cross-acceptance questions

Aspect	Acceptance question	Pass standard
Space: Types complete	Are there any gaps in the six-Type fields?	Fields complete, no "to be completed" wording
Space: numbers trustworthy	Is there a raw file behind the number?	n >= 3, statistically qualified, raw file retrievable
Space: chain traceable	Can the conclusion be traced back to the evidence?	Each of the six stations can be opened
Time, moves aligned	Does the move used match the situation?	The move-selection record matches the situation
Time, derivation complete	Are there any skipped-step conclusions?	Premise, rule, and prediction: all three present
Time, evolution leaves a trace	Has this round entered the evolution ledger?	One page per round, attribution has a name

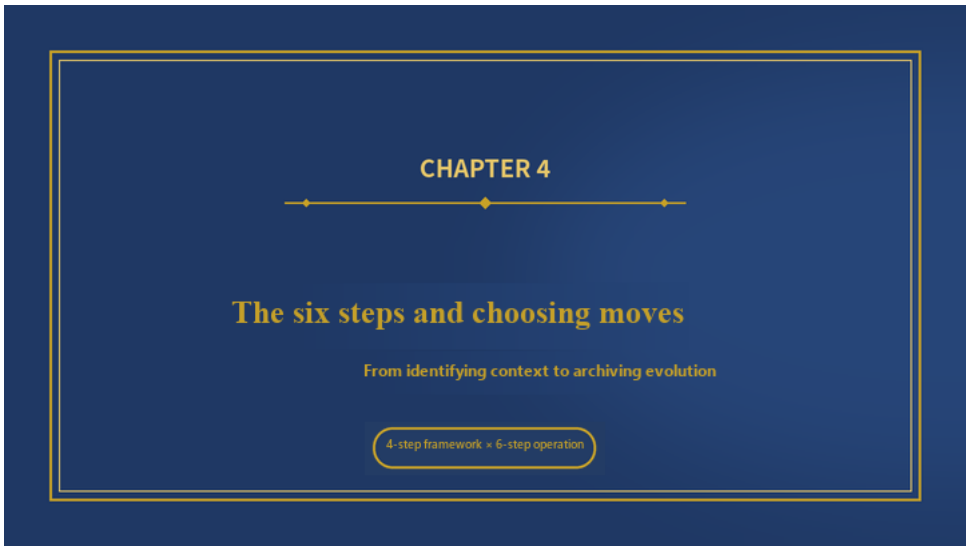
Chapter focus

Each of the four gates governs one thing: criteria govern qualification, the skeleton governs reasoning, experiment governs reality, the evidence chain governs audit - the order cannot be swapped.

The three-piece derivation "premise, rule, prediction" is the home ground of the Mind Model; a prediction must carry a tolerance interval, and a measurement must be corrected.

The six cross-acceptance questions are the last self-questioning before release - only when all six pass does the output leave this brain.

Chapter 4 Six Steps and Move Selection



From identifying the situation to archiving the evolution

A methodology only counts if it can start work. This chapter provides a process that can be followed: first use the four framework steps to identify the R&D situation, then run the six steps from situation to evolution ledger; along the way use the three-layer judgement for control and selecting, assembling, and swapping moves to deploy the Time Type.

4.1 Four Framework Steps: Identifying the R&D Situation

The four framework steps are the four questions before starting work: establish the container, characterise the property, break down the composition, set the criteria. Start before these four are answered and every later step asks the wrong question.

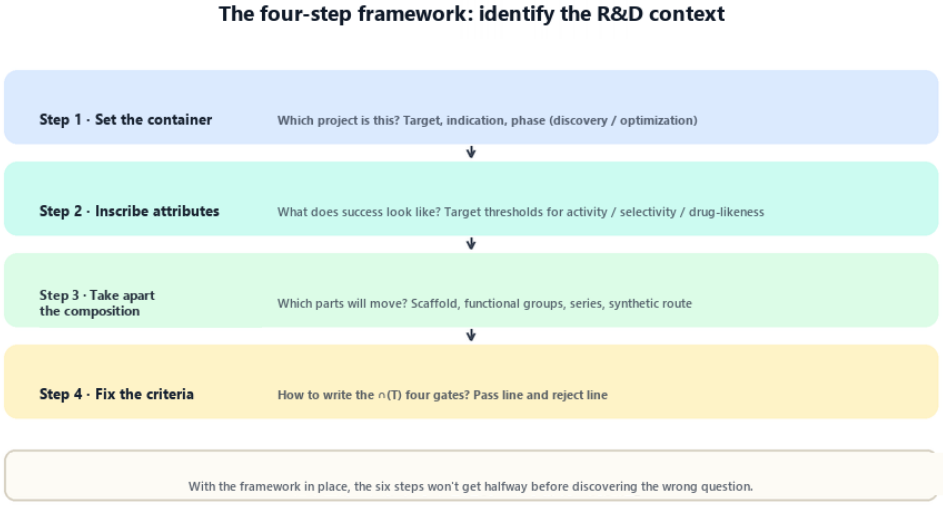


Figure 4-1 The four framework steps

Table 4-1 The four framework steps x the DX case

Step	Question	The answer of the DX case
Establish the container	Which programme and which stage is this?	Fourth-generation EGFR inhibitor (NSCLC), discovery to optimisation
Characterise the property	What does success look like?	Four-gate pass lines: activity / selectivity / druggability / ADME
Break down the composition	Which parts are to be moved?	R2 core, R1 substituent, 3-position metabolic soft spot

Step	Question	The answer of the DX case
Set the criteria	How are pass and fail judged?	Four ∩(T) gates + veto lines + rules for tracing any change of line

4.2 Six-Step Operation Card

With the framework standing, the six steps relay: identify the situation, build Types by axis, select the expansion pattern, generate the output, criteria acceptance, archive and evolve. Each step has an operation card - action, output, and commonly used moves written clearly, so a newcomer can go live by following the card.

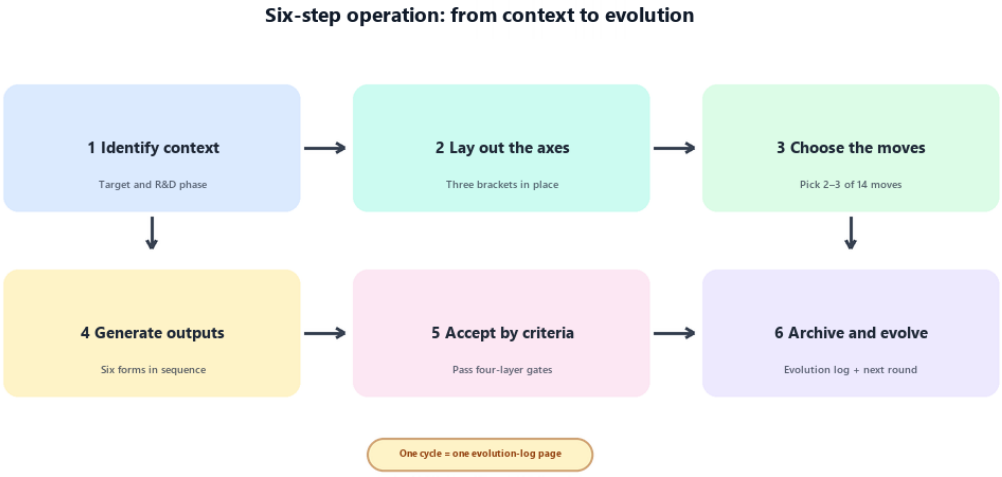


Figure 4-2 Six steps: one cycle per round

Table 4-2 The six-step operation card

Step	Action	Output	Commonly used moves
1 Identify the situation	Read the container and stage, align with the success picture	Situation card	Problem solving, analysis
2 Build Types by Axis	The three brackets in their places; the six Types on the skeleton	Six-Type field card skeleton	Design patterns
3 Select the Expansion Pattern	Pick two or three of the fourteen moves and write the exit conditions	Move selection sheet	Thinking methods
4 Generate the Output	Fill the output according to the field card and add the data	Six-Type first draft	Search, reasoning, generation
5 Criteria Acceptance	Four-gate verdict + the six cross-acceptance questions	Verdict document	Causal and reverse reasoning
6 Archive and Evolve	One page per round, criteria changes enter the library	Evolution ledger page	Continuous learning, self-evolution

4.3 Three-Layer Judgement: Is, Has, Right

Before picking up the pen for any output, pass the three-layer judgement: the first layer, container judgement, asks "is" - does it belong to this series and programme (id(C) anchoring); the second layer, structure judgement, asks "has" - is the part dependency complete (three levels of remove); the third layer, rule judgement, asks "right" - does it meet ∩(T) and the analogy rule. The order cannot be reversed: first "is", then "has", then "right".



Figure 4-3 Three-layer judgement: is, has, right

> **Example: the three-layer judgement on DX-288**
> Is: id(C) matches the DX-268 series (same container, same scaffold lineage) - identity holds.
> Yes: the pharmacophore and the core are both present, and the blocking group is third-level decoration - the structure holds.
> Right: all four gates pass + the analogy rule "pyrimidine with R1=OMe is superior" is supported by measurement - the rule holds.

4.4 Selecting, Assembling, Swapping Moves and Misuse Warnings

You need not use all fourteen moves - choosing the right two or three is more effective than deploying twelve. Select moves by situation, assemble moves by relay, swap moves by signal: when data contradicts the derivation, change the reasoning; two rounds without progress, change the search domain; criteria failing repeatedly, return to the framework.



Figure 4-4 Selecting, assembling, swapping moves

Table 4-3 Selecting, assembling, swapping moves

Situation	Selecting moves	Assembling moves	Move-swap conditions
Initial target exploration	Search and extraction	+ analysis, target card	Too few hits, so expand the search domain
SAR ranking	Causal reasoning	+ statistical reasoning, SAR Type	Correlation taken as causation, so add a counterfactual
Activity stuck	Design (creation and innovation)	+ generation, a new batch of molecules	Two rounds without progress, so refactor the scaffold

Situation	Selecting moves	Assembling moves	Move-swap conditions
ADME fails	Reverse reasoning	+ design patterns, modification plan	Three revisions fail, so change the chemical space
Verdict dispute	First-principles thinking	+ problem solving, re-evaluate the criteria	Changing a criteria line must leave a trace

Table 4-4 Six-step misuse warning points

Step	Warning point	Alert
Identify the situation	Starting before the container is established	The six Types each go their own way and do not line up
Build Types by axis	Writing the three brackets into one column	Criteria with no entry point
Select the expansion pattern	Deploying twelve moves at once	Noise in the output exceeds the signal
Generate the output	Conclusion first, derivation patched later	The evidence chain grows backwards
Criteria acceptance	If it cannot pass, change the line	Criteria versions change frequently
Archive and evolve	Reviews rely on impressions	Stepping into the same pit repeatedly

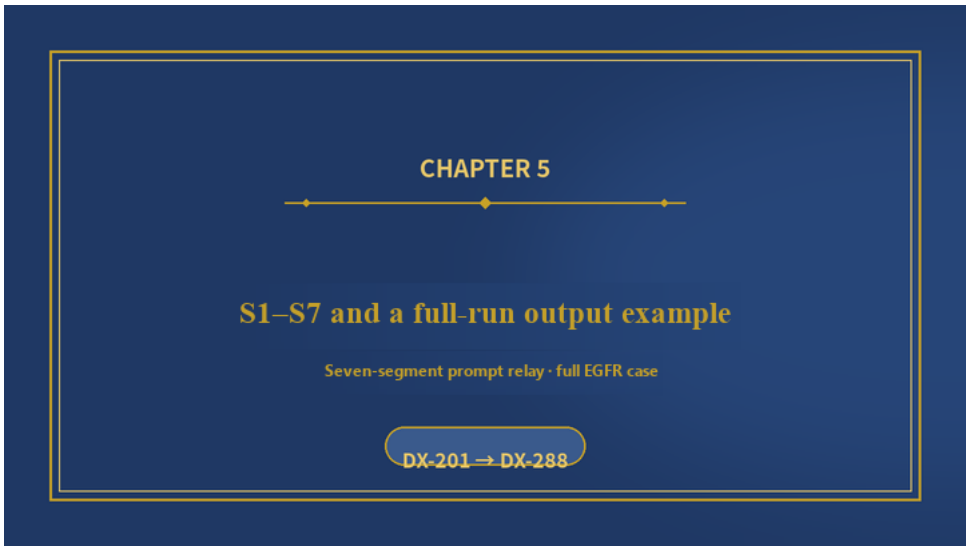
Chapter focus

The four framework steps answer four questions first, then the six steps start - asking the right question matters more than running fast.

The three-layer judgement "is, has, right" is the last self-check before picking up the pen, and the order cannot be reversed.

Moves are tools, not dogma: pick two or three, write the exit conditions, and swap if it does not work - the goal is accurate output.

Chapter 5 S1–S7 and Full Output Examples



Seven-stage relay, full DX series demonstration

This chapter gives two things you can take away directly: how to write the seven prompts, and the full output of one small-molecule targeted new drug case. The case unfolds along the DX series - from the hit DX-201 to the PCC candidate DX-288 - and the output of every Type comes with example data that are self-consistent and can be checked cell by cell.

5.1 Seven-Stage Relay: the S1–S7 Pipeline

The seven-stage relay maps to the six steps and the six Types: S1 identifies the situation and produces the situation card, S2 to S7 produce the Target Type, Hit Type, SAR Type, Optimize Type, Verdict Type, and Evolution Type in order. Each stage's output is the next stage's input; if the previous stage has not passed, the next does not start.

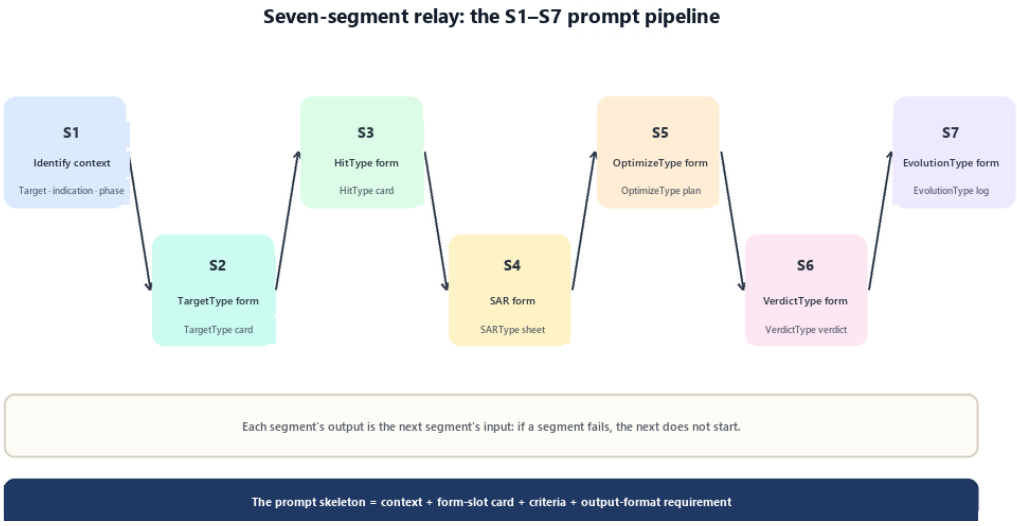


Figure 5-1 The S1-S7 seven-stage pipeline

Table 5-1 The duties and pass conditions of S1–S7

Stage	Input	Output	Pass conditions
S1	Programme background	Situation card	Container, success picture, uncertainty: all three present
S2	Situation card	Target Type card	Six fields complete, four-gate criteria in shape
S3	Target Type	Hit Type card	Screening path reproducible, noise excluded
S4	Hit Type	SAR Type table	Causal chain + counterfactual + prediction interval

Stage	Input	Output	Pass conditions
S5	SAR Type	Optimisation plan	Quantifiable target, clear exit conditions
S6	Optimisation output	Verdict document	Four gates item by item, six stations checked, sign-off complete
S7	Verdict document	Evolution ledger page	Attribution has a name, plan executable

5.2 Full Text of Prompts S1–S7

The following seven prompts can be copied and used directly. The common skeleton is the four-piece set: role, input, requirements, output format; the common discipline is "never write a number that is not supported by evidence".

S1: identify the situation (situation card)

```
You are a drug R&D planner. Produce a "situation card" from the following information.
Input: target = EGFR; indication = non-small cell lung cancer; stage = discovery to
optimisation (pre-PCC).
Requirements: (1) state the container position: programme, series, molecule;
(2) Write out the four dimensions of the success picture (activity / selectivity /
druggability / ADME);
(3) List the three uncertainties of this stage.
Output as bullet points; do not write numbers unsupported by evidence; explicitly mark
unknown items as "to be completed".
```

S2: Target Type (TargetType)

```
Continue the situation card and produce the Target Type field card.
Fields: target name, target variant, indication, mechanism, criteria ∩(T), evidence source.
Requirements: (1) write the criteria as four gates (activity / selectivity / druggability /
ADME), each with a pass line and a veto line;
(2) Every column carries an evidence source and timestamp; (3) unknown fields are
explicitly marked "to be completed" and may not be left blank.
```

S3: Hit Type (HitType)

```
Produce the Hit Type field card.
(1) Screening path: virtual library, drug-like filter, docking, in vitro primary screen,
with counts at each stage;
(2) Representative hit: structure description, IC50, LE, how it was found, noise-exclusion
record;
(3) Break down the hit with the three brackets (property / composition / container);
(4) Data are annotated with n and R2; numbers without a raw file may not be written.
```

S4: SAR Type (SARType)

```
Produce the SAR Type.
(1) List the control axes R1/R2 and the activity matrix (move only one position at a time);
(2) For the best combination, write the "change to effect" causal chain: premise, rule,
prediction;
(3) Write one remove counterfactual: what happens if the key part is taken away;
(4) Predict the next round's numerical interval, explicitly stating "prediction first,
measurement after".
```

S5: Optimize Type (OptimizeType)

```
Produce the Optimize Type plan for this round.
(1) This round's goal (quantifiable); (2) the axes locked and not moved (six axes never
regress);
(3) Design two or three knobs, with synthesis feasibility (no more than five steps);
(4) Expected interval and exit conditions;
(5) Select moves: pick two or three of the fourteen and write the reasons and the move-swap
conditions.
```

S6: Verdict Type (VerdictType)

```
Produce the verdict document.
(1) Verdict subject and id(C); (2) the four gates judged item by item: pass or fail plus
data, no bundling;
(3) Check the six stations of the evidence chain one by one; (4) one of three conclusions:
upgrade to PCC, return for revision, or shelve;
(5) Sign-off field: person, time, criteria version.
Throughout, write only evidence, not mood.
```

S7: Evolution Type (EvolutionType)

```
Produce the evolution ledger page.
(1) This round's change and the resulting change (with data); (2) attribution hypothesis +
a testable approach;
(3) Changes to the criteria library; (4) the next round's plan and move-swap conditions;
(5) One sentence to the next round's self.
```

5.3 Case, Part One: Target Type and Hit Type

Case background: a fourth-generation EGFR inhibitor programme, whose goal is to overcome the T790M/C797S cis double mutation that drives resistance to third-generation drugs, while sparing the wild type (WT) to reduce rash and diarrhoea toxicity. The data below are demonstrative, used to illustrate the output format and self-consistency.

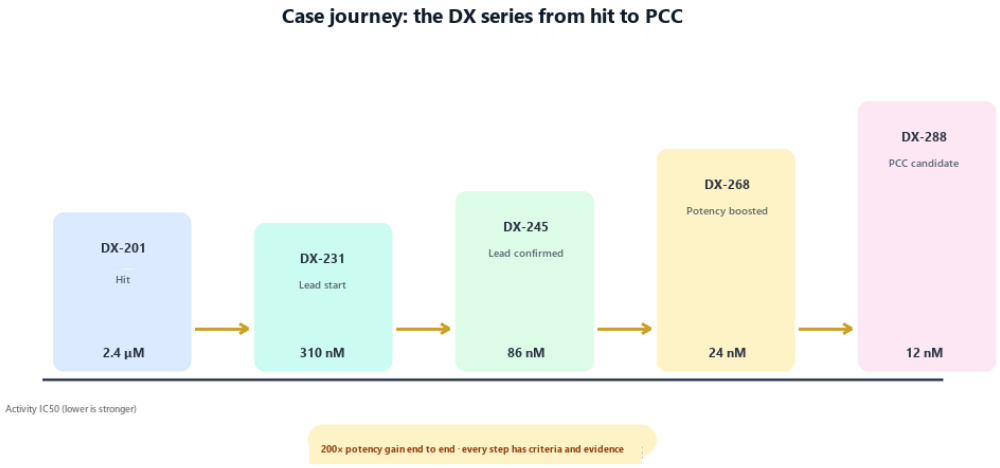


Figure 5-2 Case journey: the DX series from hit to PCC

Table 5-2 Example target card output (S2)

Field	Content
Target name	EGFR (HER1/ErbB1, tyrosine kinase receptor)
Target variant	T790M / C797S cis double mutation (the main cause of resistance to third-generation drugs)
Indication	Non-small cell lung cancer (NSCLC) resistant population
Mechanism	ATP-competitive inhibition; mutant selectivity required, sparing WT
Criteria ∩(T)	Pass lines and veto lines of the four gates (per Table 3-1, version v1.2)
Evidence source	Literature, patents, clinical resistance data (demonstration, with timestamps)

Table 5-3 Example hit card output (S3)

Field	Content
Screening path	Virtual library 50,000, drug-like filter 8,200, docking 640, in vitro 86, confirmation 12
Representative hit	DX-201: phenylpyrimidine bicyclic amide (found by fragment screening)
Initial activity	IC50 = 2.4 µM（n = 3，R^2 = 0.98）；pIC50 = 5.62
Ligand efficiency	LE = 0.31 (>= 0.3 to be shortlisted)
Noise exclusion	PAINS negative, no aggregation, no fluorescence interference

Field	Content
The three brackets dismantled	[ ] micromolar activity, soluble; { } bicyclic + amide bond; ( ) fourth-generation programme container

5.4 Case, Part Two: SAR Type and Optimize Type

The SAR stage advances with the R1 x R2 dual-axis comparison: electron-donating R1 groups raise affinity, and changing R2 to pyrimidine forms a second hydrogen bond. The activity matrix is shown in Table 5-4 - the best combination R2=pyrimidine + R1=OMe measures 24 nM (DX-268), matching the predicted interval of 20-30 nM.

Table 5-4 SAR matrix (IC50, mean of n = 3)

R1＼R2	Phenyl	Pyrimidine	Pyridine
H	2.4 μM（DX-201）	5.8 μM	3.2 μM
Me	310 nM（DX-231）	92 nM	220 nM
OMe	86 nM（DX-245）	24 nM（DX-268）	48 nM（DX-277）
F	1.1 μM	480 nM	760 nM

The optimisation stage follows: DX-268 has beautiful activity but a microsomal t1/2 of only 18 min, so the ADME gate is not passed. Following the deductive skeleton (Table 3-2), block the 3-position metabolic soft spot of the phenyl ring to obtain DX-288 - all six axes meet the target.

Table 5-5 Optimisation milestones (summary of S5 output)

Molecule	What was changed	IC50	SI	Microsomal t1/2	Determination
DX-245	R1=OMe affinity modification	86 nM	62	26 min	Lead established
DX-268	R2=pyrimidine second hydrogen bond	24 nM	85	18 min	ADME fails
DX-288	3 Block the 3-position metabolic soft spot	12 nM	118	42 min	All four gates passed

5.5 Case, Part Three: Verdict Type and Evolution Type

After three optimisation rounds, enter the verdict. The verdict document lists evidence gate by gate - no bundling, no mood; the evolution ledger records this round's changes, attributions, and criteria changes as the starting point for preclinical development.

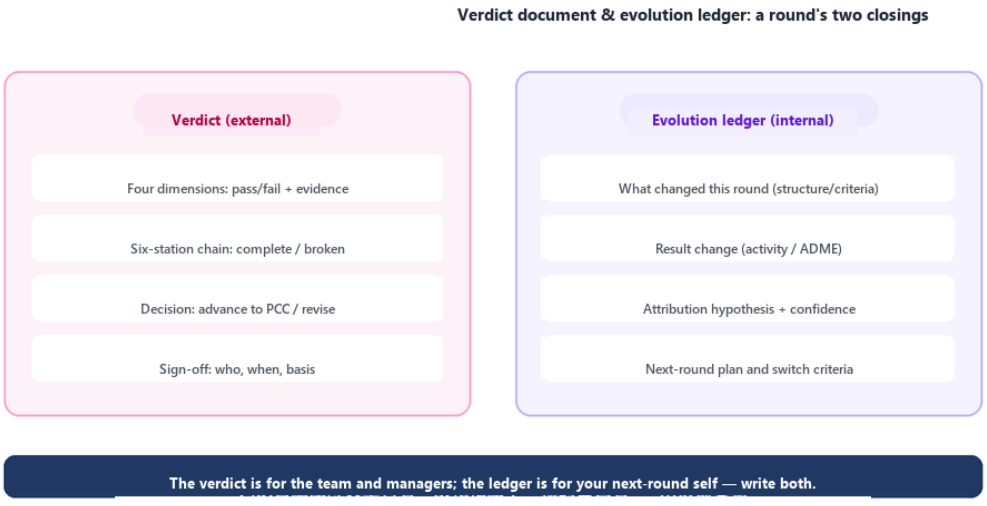


Figure 5-3 Verdict document and evolution ledger: the two closings of a round

Table 5-6 Example PCC verdict document output (S6)

Item	Content
Verdict subject	DX-288 (id(C): an upgrade within the DX-268 series)
Activity gate	IC50 = 12 nM <= 100 nM, passed (R2 = 0.99)

Item	Content
Selectivity gate	SI = 118 >= 50, passed (WT IC50 = 1.42 μM)
Druggability gate	cLogP 3.1, tPSA 92, hERG 8.6 μM, passed
ADME gate	Microsomal t1/2 = 42 min, mouse F = 48%, passed
Pharmacophore check	Mouse tumour model TGI = 85% (30 mg/kg QD x 14 days, demonstration)
Evidence chain	All six stations connected: establish Types, output, derivation, validation, verdict, archive
Conclusion	Holds: upgrade to PCC (drug candidate)
Sign-off	R&D lead, criteria version v1.2, date (demonstration)

Table 5-7 Example evolution ledger output (S7)

Field	Content
This round's changes	DX-268 to DX-288: introduce a methyl block at the 3-position of the phenyl ring (move only one position at a time)
Result change	t1/2 18→42 min · IC50 24→12 nM · hERG 4.1→8.6 μM
Attribution hypothesis	Blocking the metabolic soft spot lowers clearance (verifiable by identifying microsomal metabolites)
Criteria update	New rule "exposed 3-position = metabolic soft spot" enters the criteria library (version v1.3)
Next round's plan	Moving to preclinical development: toxicology and formulation planning; backup move: prodrug strategy
To the next round's self	Activity can deceive; metabolism cannot - look at all six axes together.

Table 5-8 Rules for using the prompts

#	Rules
1	Run only one stage at a time: open the next only after the previous passes
2	Numbers must carry n and R2; numbers without a raw file do not enter the output
3	A prediction must carry an interval, explicitly stating "prediction first, measurement after"
4	Criteria lines are set first and used afterwards; any change of line must leave a trace (person, time, reason)
5	Outputting "to be completed" is a feature, not a fault - being incomplete is not shameful; fabricating is.
6	A page of the evolution ledger must be written at the end of every round; only the next day may a new round begin

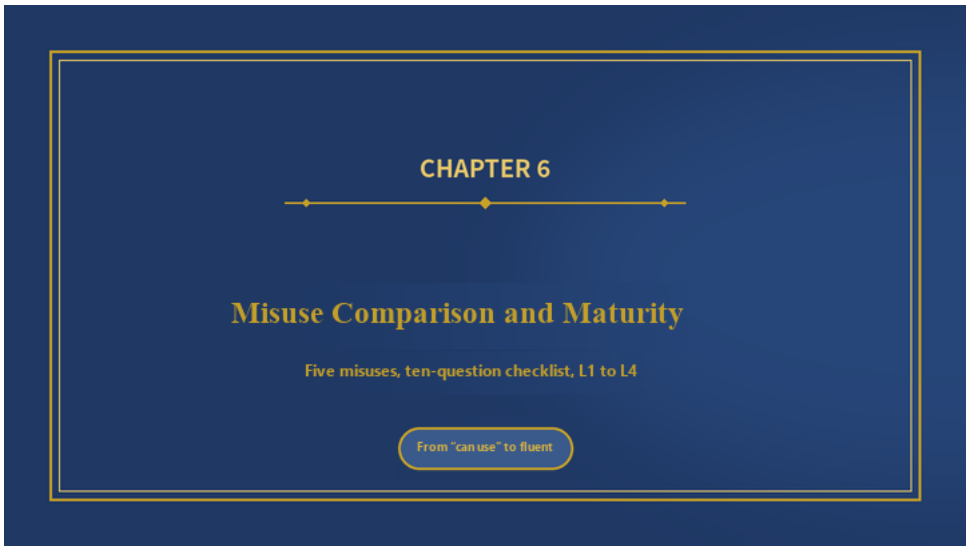
Chapter focus

The skeleton of the seven-stage relay = role + input + requirements + output format; there is only one discipline - never write a number that is not supported by evidence.

The DX series data are self-consistent throughout: 2.4 μM to 12 nM, a total of 200-fold, with criteria, derivation, and evidence at every step.

The verdict document and the evolution ledger are the two closings of a round: one tells the outside world "does it work", the other tells the inside "what was learned".

Chapter 6 Misuse Comparison and Maturity



Five misuses, ten-question checklist, L1 to L4

The method is given; the last thing is to prevent misuse. This chapter provides a comparison of five common misuses, a ten-question self-check list, four maturity levels, and a ninety-day blueprint - upgrading all the way from "can use" to "uses it fluently".

6.1 Five Misuses

Each of the five misuses crashes in its own way, but the root cause is the same: wanting to skip a gate. Whatever you skip, the cost comes back doubled later.



Figure 6-1 Five common misuses

Table 6-1 Misuse comparison table

Misuse	Symptom	Consequence	Solution
Mistaking analogy for a conclusion	Applying "the structures look similar" directly	The whole series is wiped out	Analogy counts only as a hypothesis; only passing ∩(T) counts
Looking at the activity axis only	IC50 optimised to the end	ADMET and safety all fail	Look at all six axes; no axis may regress
Loosening the criteria afterwards	If it cannot pass, change the line	Criteria lose credibility	Change of line leaves a trace + version upgrade

Misuse	Symptom	Consequence	Solution
Pushing ahead despite a broken chain	Continuing to run with missing data	Unjustified debt accumulates along the way	A broken chain means rollback and evidence reinforcement
The evolution ledger does not keep accounts	Reviews rely on impressions	Stepping into the same pit repeatedly	One page per round, mistakes never overnight

6.2 A Ten-Question Checklist

Before any output leaves the brain, self-check it with ten questions. Deliver only when all ten pass; if one sticks, go back to the corresponding gate and make up the lesson.

Table 6-2 The ten-question checklist

#	Self-check question
1	Which container does this output belong to? Does id(C) match?
2	Are there any blanks in the six-Type fields other than "to be completed"?
3	Were the criteria set first or patched afterwards? Which version number?
4	Does the prediction have an interval? Was the measurement corrected?
5	Can the key conclusions be traced station by station back to the raw data?
6	How many positions were moved this round? Was there any sneaky move of three positions at once?
7	Has any of the six axes regressed without being recorded?
8	Does the move used match the situation? Are the exit conditions written?
9	Has this round's mistake entered the evolution ledger? Does the attribution have a name?
10	Hand this output to the next round's self - can it catch it?

6.3 Maturity L1–L4

Maturity is not an honour but a stability level. Each of the four levels has a target line and common sticking points - upgrades look at evidence, not slogans.

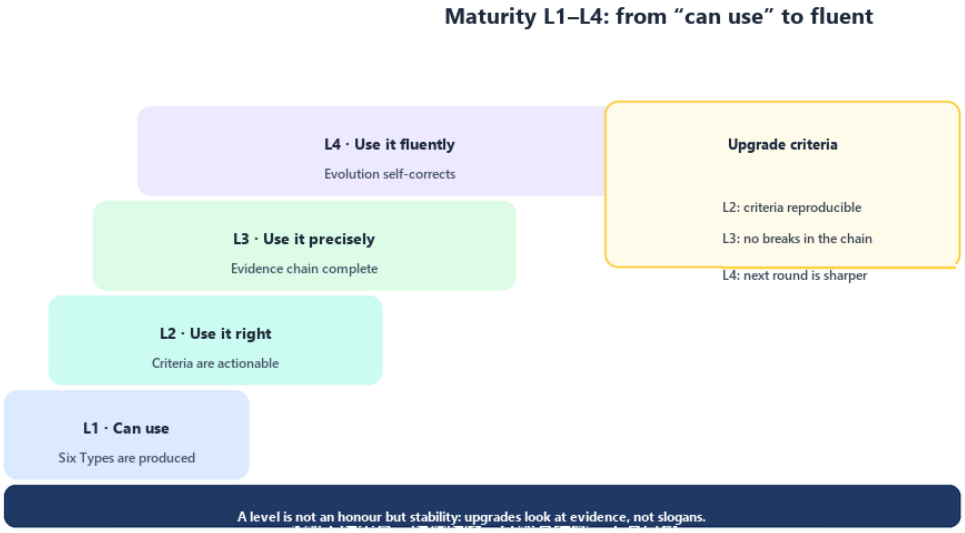


Figure 6-2 The maturity ladder

Table 6-3 The four maturity levels

Level	Feature	Target line	Common sticking points
L1: can use it	The six Types can produce	Every Type has a field card	Fields copied, criteria idling
L2: uses it correctly	The criteria can be operationalised	The four gates have pass lines and veto lines	Criteria patched afterwards, changes of line leave no trace
L3: uses it well	Evidence chain complete	Zero break points across six stations	Raw files scattered, derivation skipping steps
L4: uses it fluently	Evolution self-correction	The criteria library changes every round	Attribution unclear, hesitant about swapping moves

6.4 The Ninety-Day Adoption Blueprint

Adoption need not happen all at once: thirty days to establish the Types, thirty days to establish the chain, thirty days to establish the ledger. There is only one acceptance standard on Day 90 - when the review meeting starts, you can produce the evolution ledger instead of impressions.

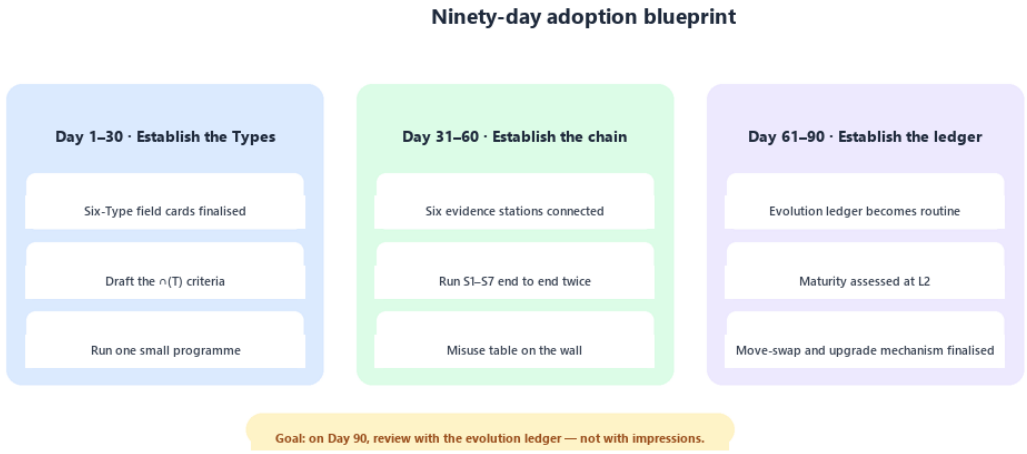


Figure 6-3 The ninety-day adoption blueprint

Table 6-4 The ninety-day blueprint

Schedule	Topic	Milestone	Acceptance
Day 1–30	Establish the Types	Six-Type field cards finalised, four-gate criteria drafted	Run one round on one small programme
Day 31–60	Establish the chain	Six stations of the evidence chain connected, two relay rounds run	Misuse table on the wall, ten questions spot-checked
Day 61–90	Establish the ledger	Evolution ledger becomes routine, criteria library activated	Maturity assessed at L2, move-swap mechanism finalised

> **Chapter focus**
> The root cause of all five misuses is "wanting to skip a gate" - and the cost comes back doubled.
> The ten-question checklist is the last line of defence before delivery; maturity upgrades look at evidence: criteria versions, chain break-point counts, and evolution ledger page counts all speak.
> Ninety days in three stages: establish the Types first, then the chain, then the ledger - on Day 90 hold the review with the evolution ledger.

Review: three takeaways



Figure R-1 Three takeaways

Takeaway one: six Types establish the output (target, hit, SAR, optimise, verdict, evolution)

Establish the Type before the output; when the Type stands, accuracy follows: Target Type, Hit Type, SAR Type, Optimize Type, Verdict Type, Evolution Type - the six Types are the minimal complete set of R&D output.

Takeaway two: four layers guarantee accuracy (criteria, derivation, validation, chain)

Accuracy is not a feeling, it is a gate: ∩(T) governs qualification, the deductive skeleton governs reasoning, the three-piece set governs reality, the six stations govern audit - only when all four layers pass does it count.

Takeaway three: the evolution ledger long run (keep accounts, attribute, swap moves)

One page per round, mistakes never overnight: keep the accounts and mistakes become criteria; swap the move and stagnation becomes progress.

Table R-1 Master table of the book's key points

Chapter	What to take away	Key tool
Chapter 1	The worldview of a reasonable twin	Type + Number, the two Types, the four-gate trailer
Chapter 2	Six-Type field cards and the three-bracket breakdown	Target, hit, SAR, optimise, verdict, evolution; three levels of remove; id(C)
Chapter 3	The four gates of accuracy	∩(T), deductive skeleton, three-piece set, six stations
Chapter 4	A workable process	Four framework steps, six steps, three-layer judgement
Chapter 5	Prompts and the full case	Seven-stage relay, DX series, usage rules
Chapter 6	Control check and the upgrade path	Misuse comparison, ten questions, L1 to L4

A final reminder

The cases and data in this book are demonstrative, used to illustrate the method; for real R&D, follow the experimental data, regulatory requirements, and professional judgement. The AI drug R&D brain is a reasonable twin of the R&D team - it amplifies expertise, it does not replace it.

Appendix A Toolbox

Three quick-reference tables: terminology, moves, notation. Use them as an index while reading and as a desk card while working.

Table A-1 Terminology quick reference

Terminology	English	One sentence
Type	Type	The complete notation of "what it is" - Type in English, not Form
Type set	Type Set	T = (F(T), ∩(T)): generation rules + intersection criterion
Intersection criterion	∩(T)	The generation rule of qualified output: it is both this Type and passes all criteria
Property	Property	[ ] Abstract set, scale axis
Composition	BOM	{ } Composition set, part dependency axis
Container	Container	( ) Spatial container: the spatial axis from programme to molecule
Identity anchor	id(C)	A stable identity that recognises itself after renaming or re-versioning
three levels of remove	remove	Recognise itself after disassembly: key, scaffold, and decoration at three levels
Move	Motion / Function	Time Type = verb = function = method
Evolution ledger	Evolution Log	One page per round: change, attribution, criteria change, plan
PCC	Preclinical Candidate	Drug candidate: the object that the Verdict Type upgrades
MPO	Multi-Parameter Optimization	Multi-parameter optimisation: six axes never regress
SAR	Structure-Activity Relationship	SAR relationship: a derivable relation of change to consequence

Table A-2 Quick reference for the fourteen moves (where they land in the Time Type)

Set	Move	Drug R&D in one sentence
One: search and extraction	Type-time function search	Finding answers in literature, patents, databases
One: search and extraction	Design search and extraction	Finding molecules with a virtual library and fragment screening
Two: reasoning and thinking	Reasoning	Causal, reverse, and statistical: solving the mystery of SAR and metabolism
Two: reasoning and thinking	Thinking methods	Look at the skeleton with first principles, look at failure in reverse
Two: reasoning and thinking	Problem solving	Which knot to untie first when multiple parameters conflict
Two: reasoning and thinking	Analysis	What the binding mode and alert data are saying
Three: design and generation	Design (creation and innovation)	Creation of new scaffolds and modification sites
Three: design and generation	Design patterns	Bioisostere and prodrug strategies applied directly
Three: design and generation	Logic generation and application	Retrosynthetic reasoning: if it cannot be made, it cannot enter the next gate
Three: design and generation	Function generation and application	Generative design: give me a batch of new candidates
Four: learning and evolution	Continuous learning	Each round's data feed back, and the criteria grow
Four: learning and evolution	Refactoring and recombination	Scaffold hopping, escaping the local optimum
Four: learning and evolution	R&D and exploration	Exploration of new targets and new indications
Four: learning and evolution	Self-evolution	The closed loop from evolution ledger to next round's plan

Table A-3 Notation quick reference

Notation	Read as	Meaning
[ ]	Square brackets	Abstract set, property (X axis)
{ }	Braces	Composition set, BOM (Y axis)

Notation	Read as	Meaning
( )	Parentheses	Container set, spatial container (Z axis)
T = (F(T), ∩(T))	Type set	Type = generation rules + intersection criterion
C = (id(C), S(C))	Container	Container = identity + structure
∩(T)	Intersection criterion	The criteria of qualified output
remove(p, M) ≠ M	remove	Take away key part p and it is no longer M
S(t+1) = g(S(t), move, event)	Evolutionary form	The progressive form of the Time Type
Space x Time	Cross	Cross-acceptance: both sides must match for it to count

The answer to drug discovery

is not more candidate molecules,

but that every molecule can say "what it is."

Target → Hit → SAR → Optimize → Verdict → Evolve — six Types, accurate output.

The AI Drug Discovery Brain is the team's reason-able research replica.

MIND AI × ONTYPOLOGY

Type = Type